SUNSTREAK

The manual · version 1.1.0



781 places · 202 countries and territories · twenty-eight legendary roads · thirty-one radio stations

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Welcome

Sunstreak is an arcade racer in the spirit of the old 16-bit road games: a low camera, a road that bends, climbs and throws you over crests, traffic to slice through, and a clock that only the checkpoints refill. The road is never the same twice. It is generated as you drive, five metres at a time, from a seed, and it carries you round the world: 781 places in 202 countries and territories, every member of the United Nations among them, the big countries state by state and province by province, through the seasons and their festivals, across the seas in glass tunnels on the sea floor and over the passes in numbered hairpins.

Everything you see and hear is made by the program itself: the low-poly world, the sky, the stroke font, the engines with four, eight or twelve cylinders, and thirty-one radio stations that compose their tracks as you listen.

This manual tells you how to drive, what the screen shows, what the modes are, and what you will meet on the road. The pictures are taken by the game itself.



The title screen. Left and right on ENDLESS RUN choose between resuming the world tour where you stopped and starting a new one.

Getting started

INSTALLING

- **Windows**: unpack `Sunstreak-<version>-windows-x64.zip` anywhere and start `Sunstreak.exe`. The game starts in a window; F11 or Alt+Enter switches to fullscreen.
- **Linux**: unpack `Sunstreak-<version>-linux-x86_64.tar.gz` and start `Sunstreak`. The program needs only the C library; OpenGL and X11 are loaded when it runs.
- **Steam Deck**: copy the Linux program onto the Deck and add it to Steam as a non-Steam game. It runs in Gaming Mode and starts fullscreen, with the BALANCED profile and the text a size larger.

Your own music can come along: put `.ogg`, `.mp3`, `.flac`, `.wav`, `.xm` or `.mod` files into the `music` folder beside the game, and they play as the station MY MUSIC.

THE TITLE SCREEN

Entry	What it does
ENDLESS RUN	The world tour against the clock, from where the last run stopped (or a NEW TOUR)
TRAVEL	The world tour without a clock, for the places, the passport and the postcards
DAILY CHALLENGE	Today's track, the same all day, with a twist
RALLIES	Ten long-distance routes and the journeys between two places
CHAMPIONSHIP	Five races against eleven rivals
TRACK LAB	Build a track from its settings, or type in a code
GARAGE	Your car, its paint, rims, decals and number plate
PASSPORT	The map of the tour, stamps, regions, postcards, album and awards
RECORDS	The top ten, your kilometres, titles and progress
SETTINGS	Picture, sound, assists, the world's weather and seasons

Controls

	Keyboard	Gamepad, Steam Deck
Accelerate	Up / W	RT or A
Brake; held at a standstill: reverse	Down / S	LT or X
Steer	Left, Right / A, D	left stick, d-pad
Boost	Space / Shift	B
Gear down / up (manual gears)	Q / E	LB / RB
Camera: chase, far, bumper, cockpit	C	Y
Look back (hold)	B	right stick pressed in
Flash the headlights: slower traffic ahead makes way	F	left stick pressed in
Horn: animals leave the road	H	d-pad up
Radio station / next track	M / N	View
Pause	Esc / P	Start
Window / fullscreen	F11 or Alt+Enter	
Performance overlay	F3	
Read the data files again	F5	
Landmark viewer	F6	

The gamepad hints on screen show the buttons as glyphs (on the Steam Deck its own), with a keyboard they show the keys. Gamepads rumble on hits, crashes, landings, boosts and gear changes and buzz on kerbs and gravel; VIBRATION in the settings turns it down or off.

The screen



Driving through Tuscany.

- **Top left**: the score, the kilometres driven and the height above the sea (ALT; in feet with MPH), and the stage line: the stage, the place, its region and its country.
- **Top middle**: the clock. When it reaches zero, the run ends; in TRAVEL it shows the time on the road instead.
- **Top right**: the best score so far, your overtakes and, after near misses, the combo.
- **Left**: messages as they come (a front ahead, a hitchhiker waiting, a duel on), the distance to the next checkpoint, and the minimap with the road ahead.
- **Bottom right**: the gear, the speed, the rev band and the boost tank.
- **Middle**: big news: a checkpoint and its bonus, a new stage, a speed camera, a fine.

C changes the camera: chase, far, bumper and cockpit. The cockpit puts you in the driver's seat, on the left, or on the right where they keep left, with live dials and what the country's drivers hang from the mirror.



The cockpit at night in Tokyo, the beckoning cat on the dashboard.

Driving

THE CAR

Accelerate, brake and steer. In a bend the road pushes the car outwards, the more the faster you go: steer against it, and brake before the tight ones. The minimap and the chevron boards show what is coming; the yellow diamonds and triangles warn of crests, and a crest taken fast throws the car into the air.

Stuck in the verge or against a rock at a hairpin? Hold the brake at a standstill and the gear shows **R**: the brake now backs the car up (up to 20 km/h, as far as 150 m), and the throttle stops it and takes it forward again. Off the road the car is slower, but it can always crawl back.

The **boost** (Space, B on a gamepad) adds speed while the tank lasts. The tank fills with **near misses**: pass traffic close and fast. Every near miss also raises the **combo**, which multiplies what the next ones are worth.

Take a bend near the limit and the car **drifts**: its nose turns into the bend, the tail swings out, the outer rear tyre smokes and leaves its mark on the asphalt, and at the bend's end the body swings back. A drift of 30 m or more earns 10 points a metre (times the combo) and raises the combo. Beside a guard rail or close to traffic the car turns less, so it never looks as if it hit what it does not. DRIFT in the settings: OFF; LOOK ONLY, the look and the points, the car driving exactly as without it; or FULL, where in a drift the car answers the wheel a little more and its sideways motion a little later, and the drift fills the boost tank a little.

Close behind a car in your lane at speed you are in its **slipstream**: the air holds you back less, you pull a little harder, the wind falls quieter and streams past. Overtake straight out of it for a SLIPSTREAM PASS (+300, times the combo).

With **ONCOMING TRAFFIC** on in the settings (LIGHT, NORMAL or DENSE; it is off at first), a road of two or four lanes is a road of two directions, as its markings show it: the half across the middle, the left one or the right one where they drive on the left, carries what comes the other way. The traffic of your direction keeps to its half, so on a road of two lanes there is no going round a slower car without going out into the other lane, and that only when nothing comes. What comes the other way flashes its lights and sounds its horn at a car in its lane; it keeps to its edge of the road to leave room, and so does the car you squeeze past, so there is a way through the middle. Missing it close and fast is a CLOSE CALL (+1000, times the combo, and a good gulp for the boost tank); a red arrow at the top of the picture blinks when something comes at you in your lane; meeting it is a head-on crash. An overtake on such a road is worth more: one and a quarter times on LIGHT, one and a half on NORMAL, twice on DENSE, where the oncoming traffic overtakes too, out in your half while it sees the way clear, and back in its own before it meets you. A lorry or a bus going by shoves the car a little with its blast. The line down the middle is solid where nobody sees far enough to overtake: round tight bends, over crests, and on four-lane roads. Roads of three lanes stay one carriageway, a fork's two carriageways both lead away, and at roadworks the other direction waits until yours is through. The autopilot and STEER ASSIST overtake only with room for it, and not over a solid line.



Oncoming traffic at night in the Black Forest.

In the rain the lamps of the traffic lie mirrored in the wet road, and the traffic throws up spray (slush in the snow, dust on gravel and sand).

Flash the headlights (F, or the left stick pressed in) and slower traffic ahead in your lane pulls over to the slow side for a few seconds, indicating; where the lines are a suggestion, only some of it does. The traffic itself indicates before it changes lanes (again: not everywhere), and its brake lights show when it slows.

The **horn** (H, or up on the d-pad) sounds as the country's horns do, one note in Europe, a deep long one in North America, a little tune in South and Southeast Asia; animals in the road stop dawdling and leave it, and where the lines are a suggestion somebody answers. Red arrows at the picture's edges warn of something close beside the car or coming up fast from behind. After long hard braking the brakes glow in the wheels.

The car shows where it has been: dust and mud gather on it off the road and on gravel and sand, slush in the snow, and the rain washes it clean again. On snow, sand, gravel and the verge every wheel leaves its rut; on the legendary roads the game keeps your longest drift.

GEARS in the settings switches between the automatic and a manual gearbox (Q and E, or the shoulder buttons).

THE ROAD

Asphalt holds best. Gravel, dirt and sand hold less and throw up dust; the Gibb River Road, the Yungas, the Sani Pass, the Road of Bones and the Sahara have no asphalt at all, and the country's road quality decides how much gravel, how many patches and potholes a road has elsewhere. Wet roads, snow and sand hold less; in a downpour, puddles gather, and over 100 km/h the tyres float on them: AQUAPLANING, hardly any grip until they find the road again. Cobbles shake.

Hitting traffic, a sign or a stand costs speed; a crash costs more. The barriers along bridges, tunnels and galleries keep you on the road.

HELP

STEER ASSIST (light or strong) steers where your hands are idle, keeps the road and goes round the traffic; BRAKE ASSIST brakes for bends, obstacles and animals taken too fast. DIFFICULTY (easy, normal, hard) scales the grip, the clock, the traffic and the rivals. The handling itself is in the file `tuning.ini` beside the settings, written with comments on the first start: change a value and restart, delete the file for the defaults.

The clock and the score

The clock starts at 55 seconds. Every checkpoint (one every 2.5 km) adds time, a little less each stage; where the road climbs a pass in hairpins, the checkpoint counts the slow bends in. The run ends when the clock reaches zero, and the results show the track code to drive the same road again.

The score grows with speed and distance, overtakes, near misses, drifts and slipstream passes (times the combo), and the bonuses of the road:

What	Bonus
A checkpoint	+1000 and its time
Getting away from the police (400 m ahead)	+2500
Letting an ambulance or a fire engine through	+1000
A hitchhiker or a courier job delivered	+3000 and five seconds
Beating a fast train to the end of its viaduct	+4000 and five seconds
Beating a local hero to the next checkpoint	+5000 and five seconds
Caught by the police	five seconds off
A speed camera, passed too fast	two seconds off

In TRAVEL there is no clock, and the fines are taken in points.

The modes

ENDLESS RUN

The world tour against the clock. It continues where the last endless run stopped; NEW TOUR (left and right on the title) starts again from the beginning. SAVE AND QUIT in the pause menu keeps the run as it stands, the place on the road, the clock and the score; RESUME on the title drives on from there. With SCENERY set to RANDOM TOUR, the road forks after every stage to two countries drawn at random, each in another part of the world and never one the round has been to; across the sea between landmasses on the sea floor, a ferry or the car train. The round goes on from run to run, the stage line counts it (37 / 202 COUNTRIES), and once every country is done the next round begins. With SCENERY set to ALL MIXED, every stage is another biome, the twelve fantasy biomes included: Night Metro, Storm Marsh, Neon Horizon, Neon Rain, Dawn Expressway, Lunar Highway (with low gravity), Red Planet, Crystal Desert, Fungal Glade, Cloud Highway, Pumpkin Hollow and Total Eclipse.

TRAVEL

The world tour without a clock, on and on from where the last tour stopped: for the places, the passport and the postcards. Speed cameras and the police still fine you, in points.

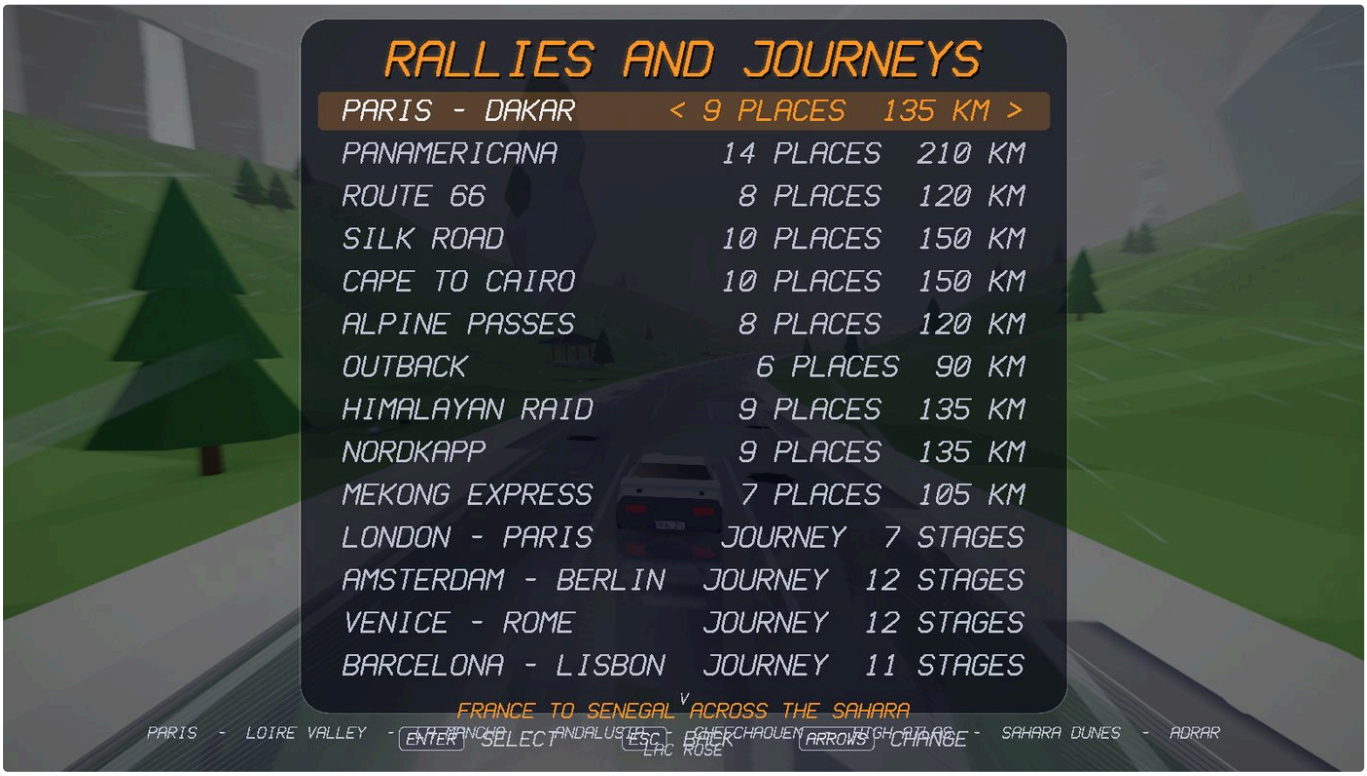
DAILY CHALLENGE

Today's track, the same for everybody all day: a world tour from a seed of the date, with a twist such as rush hour, roadworks everywhere or all bends. The best score of the day is kept.

RALLIES AND JOURNEYS

Ten long-distance routes driven in one go against a clock that counts up, place after place: Paris - Dakar, the Panamericana from Alaska to Patagonia, Route 66, the Silk Road, Cape to Cairo, the Alpine passes, the Outback, a Himalayan raid, the Nordkapp and the Mekong Express. Each finished rally keeps its best time.

The journeys run from one place of the tour to another: London - Paris, Venice - Rome, Osaka - Tokyo, up to Lisbon - Tokyo, half the world. Here the forks matter: before each, the screen shows how many stages the goal is either way, the shorter in gold; the other may be a detour or a shortcut. Each journey keeps its best time and its ghost.



The rallies and journeys.

CHAMPIONSHIP

Five races on generated tracks against eleven rivals, from the back of the grid. Points go 10-8-6-5-4-3-2-1 per race, and the standings come after each. The rivals draft in your slipstream, and you in theirs.

TRACK LAB

Every track is a combination of settings: the mode (time attack or a race against the eleven), the scenery (the world tour, all mixed, or one place you have driven through), curves, sharpness, hills, traffic and obstacles (0 to 9, or AUTO), the length (2.5 to 37.5 km, or endless in time attack) and the seed. Each combination is a code of fifteen characters, such as `K3ZQ8-M2TVX-41P0R`, with a check symbol against typos. CODE types one in; the results of every run show its code. Send a code to a friend, and you drive the same road. The best drive on each code is kept as a ghost car that races along the next time.



The Track Lab.

REPLAY AND PHOTO MODE

After a run, WATCH REPLAY plays the last minute again from roadside cameras, a helicopter and the chase camera. PHOTO MODE in the pause menu stops the game and lets the camera roam round the car; a photo is saved in the `photos` folder beside the settings and becomes the place's postcard.

The world tour

STAGES AND PLACES

The tour runs east from Ireland, round the world and back: the British Isles, France, Germany and the Alps, Italy and Iberia, Africa, Arabia and the Levant, Turkey and the Balkans, Eastern Europe, Central Asia, India and the Himalaya, Southeast and East Asia, Australia and the Pacific, South and Central America, Mexico, the United States and Canada, Russia, Scandinavia, Iceland. Every 15 km a new stage begins with a new place, and the stage line names it with its region and country: HILL COUNTRY - TEXAS - USA.

Each place greets you with its name on a sign in the national style. Where the country writes in its own script, the sign is bilingual, the name in that script above the Latin one: Cyrillic, Greek, Georgian, Armenian, Hebrew, Arabic and Persian, the scripts of India state by state, Sinhala, Thaana, Thai, Lao, Khmer, Burmese, Ge'ez, Chinese, Japanese and Korean. Where a state or province begins, a board says so; where the country changes, there is a border post with both flags.



The place sign of Tokyo, in Japanese and in Latin letters.

THE FORKS

At most borders the road forks. Almost two kilometres ahead, two boards on the screen show where each side goes (the place, its region and country, the distance counting down) and light the side you are on. Gantries announce it on the road, the road widens into two carriageways, and a barrier with a crash cushion divides them: the side you are on at its nose decides. The next stage is the place you chose.



Keep left for Mexico City, right for Morelos.

ACROSS THE SEAS

Where two places lie on different landmasses that no land joins (Africa and Asia meet at Sinai, the Americas at Panama), a stage on the sea floor lies between them, a third as long as one on land: a glass tunnel of steel ribs, the light of the sea instead of the sun, shoals, sharks, whales, mantas and jellyfish swimming along the glass, a wreck or a research station on the floor. Forty-two seas have their own floor and their own life. Seven seas are crossed aboard instead: the Channel on the car train, the Irish Sea, the Baltic, the Strait of Gibraltar, the Aegean, the Balearic Sea and the Java Sea on a ferry. You drive aboard yourself, the clock stands still on the crossing, and you drive off at the far quay.



In the glass tunnel under the North Atlantic.



On the ferry across the Irish Sea.

LEGENDARY ROADS

Twenty-eight of the places are legendary roads. A banner announces them when they begin, the game keeps your best time over each from end to end, and each has its signature. The passes climb in hairpins, numbered on the outside and counting down to the top, where the pass's name and height stand on a board, and the road is as high as the board says: coming from far below, a pass begins with a long climb up its valley to the foot of the hairpins. Everywhere the road lies as high as the place's roads really do: it climbs and falls between the places, from the Dead Sea 400 m below the sea to the Altiplano at almost 4000 m and the Himalaya above 5000. All the legs of a climb lie on one mountainside, and the legs below stay in sight. Other high mountains climb in hairpins now and then too. The Atlantic Road has its humped bridges over the sea that seem to end in the air.

Road	Country	What it is	Hairpins	Top
Applecross Pass	Scotland	The pass of the cattle	6	Bealach na Ba 626 m
Atlantic Road	Norway	8 bridges from islet to islet		
Big Sur	USA	Highway 1 on the cliffs of Big Sur		
Chapmans Peak	South Africa	114 curves on the cliffs		
Col de Turini	France	The night stage of the Monte Carlo Rally	16	Col de Turini 1607 m
Dalton Highway	USA	666 km to the Arctic Ocean		
Furka Pass	Switzerland	Up to 2429 m past the Rhone Glacier	12	Furkapass 2429 m
Going to the Sun Road	USA	Over the Continental Divide		Logan Pass 2026 m
Great Ocean Road	Australia	243 km along the Southern Ocean		
Grossglockner	Austria	36 hairpins up to the Hochtor	36	Hochtor 2504 m
Irohazaka	Japan	48 hairpins named after the Iroha poem	24	Chuzenji 1269 m
Karakoram Highway	Pakistan	Up to 4693 m at the Khunjerab Pass		
Manali Highway	India	The 21 Gata Loops	21	Nakee La 4739 m
Million Dollar Highway	USA	No guard rails at 3358 m	8	Red Mountain Pass 3358 m
Nordschleife	Germany	20.8 km and 73 corners of the Green Hell		
Nullarbor Plain	Australia	146.6 km dead straight		
Pikes Peak	USA	156 turns up to 4302 m	16	Pikes Peak 4302 m
Ring Road	Iceland	1322 km round Iceland		
Road of Bones	Russia	2000 km through the coldest place on Earth		

Road to Hana	USA	620 curves and 59 bridges		
Route 66	USA	The Mother Road		
Sani Pass	Lesotho	Gravel hairpins up to Lesotho	14	Sani Top 2876 m
Stelvio Pass	Italy	48 hairpins up to 2757 m	48	Passo dello Stelvio 2757 m
Tail of the Dragon	USA	318 curves in 11 miles		
Tianmen Mountain	China	99 bends up to Heaven's Gate	33	Tianmen 1300 m
Transfagarasan	Romania	90 km across the Fagaras Mountains	27	Balea Lac 2042 m
Trollstigen	Norway	11 hairpins beside the Stigfossen	11	Trollstigen 858 m
Yungas Road	Bolivia	The Death Road		



Tornante 25 on the way up the Stelvio.



The hairpins of the Grossglockner from above.



The top of the Stelvio, 2757 m.



A humped bridge of the Atlantic Road.

CITIES, VILLAGES AND THE ROADSIDE

Cities stand on plots in rows some 300 m back from the road, cross streets between their blocks; in the country the houses gather in villages of two or three rows, with lone farms between. The houses are the region's own: canal houses in the Low Countries, machiya in Japan, white Cycladic cubes, falu-red cottages in Scandinavia, izbas in Russia, azulejo facades in Portugal, stilt houses in Southeast Asia, Queenslanders, the tower houses of Yemen and many more.



New York at night.



A village in the Black Forest.

Where a region sells by the road, its stands stand there, built its way, as many as it has and in their season: honesty boxes at farm gates, strawberry kiosks in early summer, pumpkins in an American autumn, melon cages in Russia, chai stalls and dhabas in India, fruit stalls on bamboo in Thailand, unmanned vegetable shelves and vending machines in Japan, charcoal sacks in East Africa, fruit carts under parasols in Latin America, sixty kinds in all. Wind farms turn where the wind blows; there are landmarks once per stage (Big Ben, the Colosseum, the Statue of Liberty, the Taj Mahal, Machu Picchu and over two hundred more), and the Hill of Crosses, the towers of Svaneti, Bhutan's dzongs, Bali's meru towers, funfairs, Dutch lift bridges, truck stops and refineries along the way.



Vending machines by a road in Hokkaido.

BRIDGES, TUNNELS AND GALLERIES

Bridges are built the way the region builds them: concrete beams, tall viaducts over Alpine gorges, stone arches, steel trusses, suspension bridges over the Bosphorus and in New York, cable-stayed spans, covered wooden bridges, causeways over the sea. Tunnels go through the hills; on Alpine roads, avalanche galleries roof the road against the mountain, open to the valley.

Life on the road

TRAFFIC

The traffic drives the country's way, and as much of it as the country has: swarms of scooters in Vietnam, Indonesia and India; cars weaving across the lines and honking in India, Egypt or Nigeria; tractors on the country roads of Europe; road trains in the outback. The vehicles are the region's own, with the country's number plates: yellow cabs and big rigs in North America, black cabs and double-deckers in Britain, painted chicken buses in Latin America, auto rickshaws and decorated trucks in South Asia, jeepneys and tuk-tuks in Southeast Asia, kei trucks in Japan, matatus in Africa. Where they drive on the left, the slow traffic keeps left and you start in the left lane. Out in the country the animals of the place cross the road: cattle, sheep, kangaroos, moose, camels, elephants, llamas, reindeer and more.

Every few kilometres an ambulance or a fire engine of the country comes along on a run, lights flashing and siren going. The traffic ahead of it pulls over; keep out of its way when it overtakes and you earn +1000. Sirens and horns are Doppler-shifted: higher as they come nearer, lower as they go.

THE POLICE

On about one stage in three, a patrol car of the country's police waits at the roadside, in its own livery. Pass it too fast and it gives chase with its lights and siren. Get 400 m ahead and you have escaped (+2500); let it pull up beside you and the fine is five seconds. Hold B (the right stick pressed in) to look back. POLICE in the settings turns them off. Speed cameras flash in Germany, Austria, Switzerland, France, the Low Countries, Italy and Britain; the limit stands before them.

HITCHHIKERS AND COURIER JOBS

Now and then someone waits by the road, thumb out. Slow down beside them and they get in, with a hello in their language; take them to the next place before their patience runs out (+3000 and five seconds, and their thanks). Sometimes it is a courier job instead:

- a courier with a fragile parcel for the next place: hit nothing on the way, or it breaks;
- a paramedic with an organ for the hospital at the next checkpoint: hurry;
- a police officer whose car was just stolen: catch the thief by staying close beside or behind it for a moment before it reaches the next checkpoint.



A hitchhiker ahead.

LOCAL HEROES

Now and then a local driver pulls up beside you before a checkpoint, in a car of your class, and challenges you to the next one in their language. Beat them there for +5000 and five seconds (and a grumble); let them win and they taunt you.



Ewan of the Isle of Skye wants a race.

FAST TRAINS

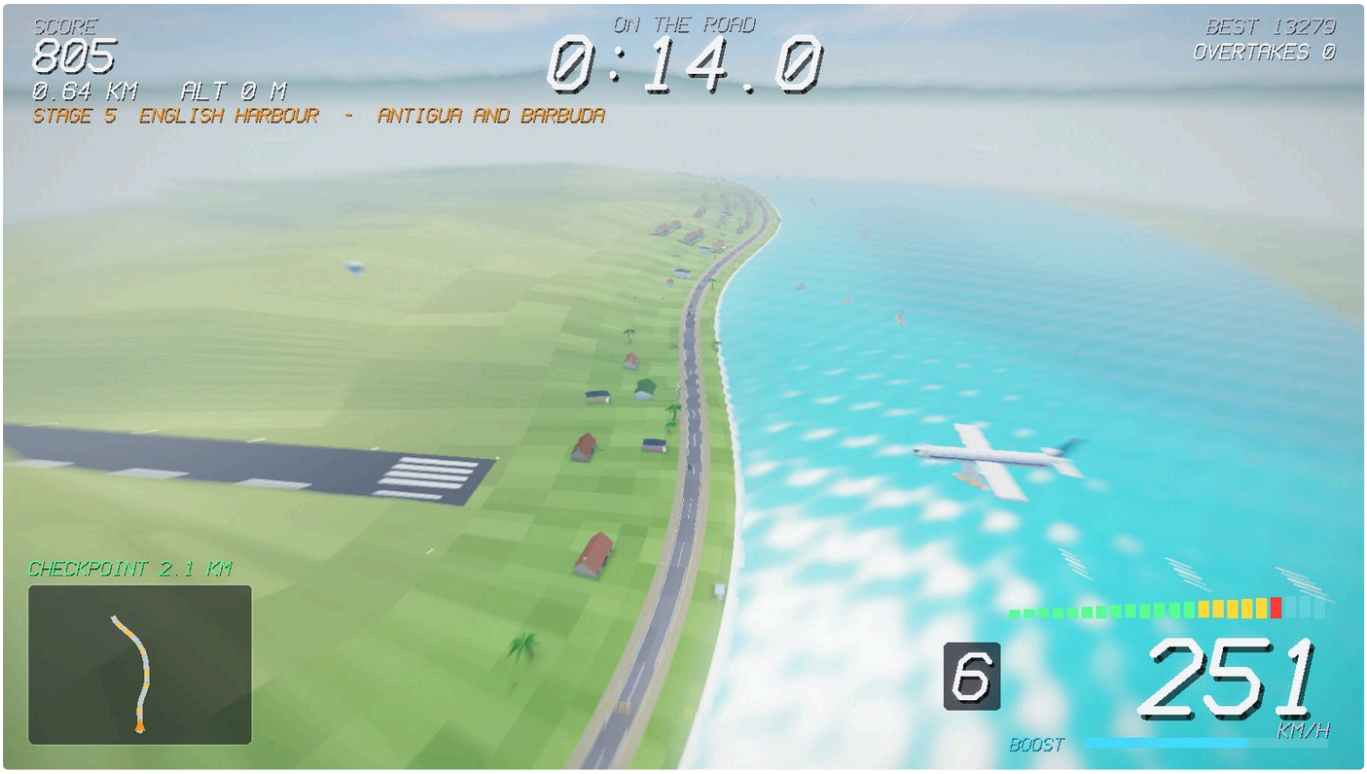
In half the stages of countries with a fast train (the Shinkansen, the TGV, the ICE, the AVE, the Frecciarossa, the KTX, the Indian Pacific and ten more) a viaduct runs beside the road for a few kilometres, out of a stone portal and into another, and the train races you along it. Pass its nose before the stretch ends for +4000 and five seconds.



Racing a train along its viaduct.

SOMETHING TO SEE

Now and then the sky has something to show: an airliner coming in to land low across the road near the big airports and over the Caribbean islands, its runway beside the road; fighter jets in the deserts of the American west and the Emirates; hot-air balloons over Cappadocia, Bagan, Luxor and the Serengeti; a container ship on the Panama Canal; an airship over southern Germany.



An airliner coming in over the sea of Antigua, its runway beside the road.

Weather, light and seasons

WEATHER

Weather fronts pass over the tour: showers, fog banks and thunderstorms in temperate places, downpours in the tropics, snow showers in the cold and the mountains, sandstorms in the deserts. They are announced a little ahead and last a kilometre or two. In the rain the road turns into a dark mirror with the cars, the trees and the street lamps standing in it. In South and Southeast Asia every other tropical front is the monsoon, with stretches where the road is flooded from edge to edge. In the snow, grey slush lies in blotches and sprays up behind the car. Fog lies low in the valleys at dawn and dusk; over the deserts dust devils whirl, and where it is hot the air shimmers over the asphalt. WEATHER in the settings chooses between the fronts and each place's own weather.



Rain in Glencoe, and a speed camera.



The monsoon in Kerala.

LIGHT

TIME OF DAY keeps each place's own light by default. TOUR CLOCK lets the sun follow the time zones: morning wherever the run starts, an hour more every ten kilometres, and the longitude of each place on top, so east of you it is already evening. DAWN, DAY, DUSK and NIGHT fix the hour.

SEASONS AND FESTIVALS

SEASONS follows today's date (AS TODAY), turns a year every 400 km (AS YOU TRAVEL), or stays OFF. In autumn the broadleaf trees, hedges and vines turn orange, red and gold and leaves blow across the road; in winter snow lies on the land and on everything that faces up, and it snows; in spring the green is fresh and some trees blossom. When the seasons come follows each place's real climate: snow lies where the air stayed below freezing, so Patagonia is white in July, while the tropics and the deserts keep their one look. On the roads of Aomori, Logan Pass, the Grossglockner, the Furka, the Stelvio and Trollstigen, walls of snow line the road in winter and spring.



Between the walls of snow in Aomori.

On their dates, the festivals come:

Festival	When	Where
The festival of colours	01.03. to 20.03.	India, Nepal
Samba in the streets	06.02. to 04.03.	Brazil
Red lanterns for the new year	21.01. to 20.02.	China, Taiwan, Vietnam, Singapore, Malaysia, South Korea
The festival of lights	20.10. to 15.11.	India, Nepal, Sri Lanka
Mulled wine and fairy lights	24.11. to 26.12.	Germany, Austria, Switzerland, Czechia, France/Grand Est
Beer tents and blue and white	17.09. to 04.10.	Germany/Bavaria
Happy new year	31.12. to 01.01.	Everywhere
The aurora is out	15.09. to 01.04.	North of 60°



Lights over the road on a December night on the Romantic Road.

THE AIR

The air is as clean or as dirty as the place's pollution index says: over Delhi, Kathmandu, Lagos or Cairo the haze thickens and browns, and litter lies by the road, while Helsinki and Reykjavik stay clear.

The radio

Thirty-one generated stations, every track composed from its own seed and named: its tempo, key and scale (from major and minor to Hijaz, Bhairav, hirajoshi and pelog), its chord progression, two melodies, a lead instrument and an arrangement of 32 to 64 bars with fills and risers.

- **Genres**: Sunset FM (synthwave), Turbo House, Amiga Techno, Trance Gate, Jungle Pressure (drum and bass), 8-Bit Racer (chiptune), Fusion Drive, Big Beat, Route 66 Radio and Lo-Fi Cruise.
- **Regions**: City Pop FM, Koto Wave, Bollywood Express, Desert Caravan, Afro Highway, Samba Avenida, Tango Turbo, Andes Wind, Caribbean Steel, Mariachi Motor, Celtic Coast, Balkan Brass, Gamelan Drive, Outback Drone, Steppe Rider, Nordic Night, K-Pop Neon, Fado Noite, Schlager Express, Polka Parade and Afrobeat Lagos.

In AUTO, the station follows the place: the countryside plays its region, cities and motorways their genre, and driving into a new region the station sings its jingle and says where it plays from. M switches the station, N skips to the next track. With little time left, or on the last kilometre of a race, a driving layer joins in, and checkpoint, stage and finish cues land on the next beat.

The garage



The garage.

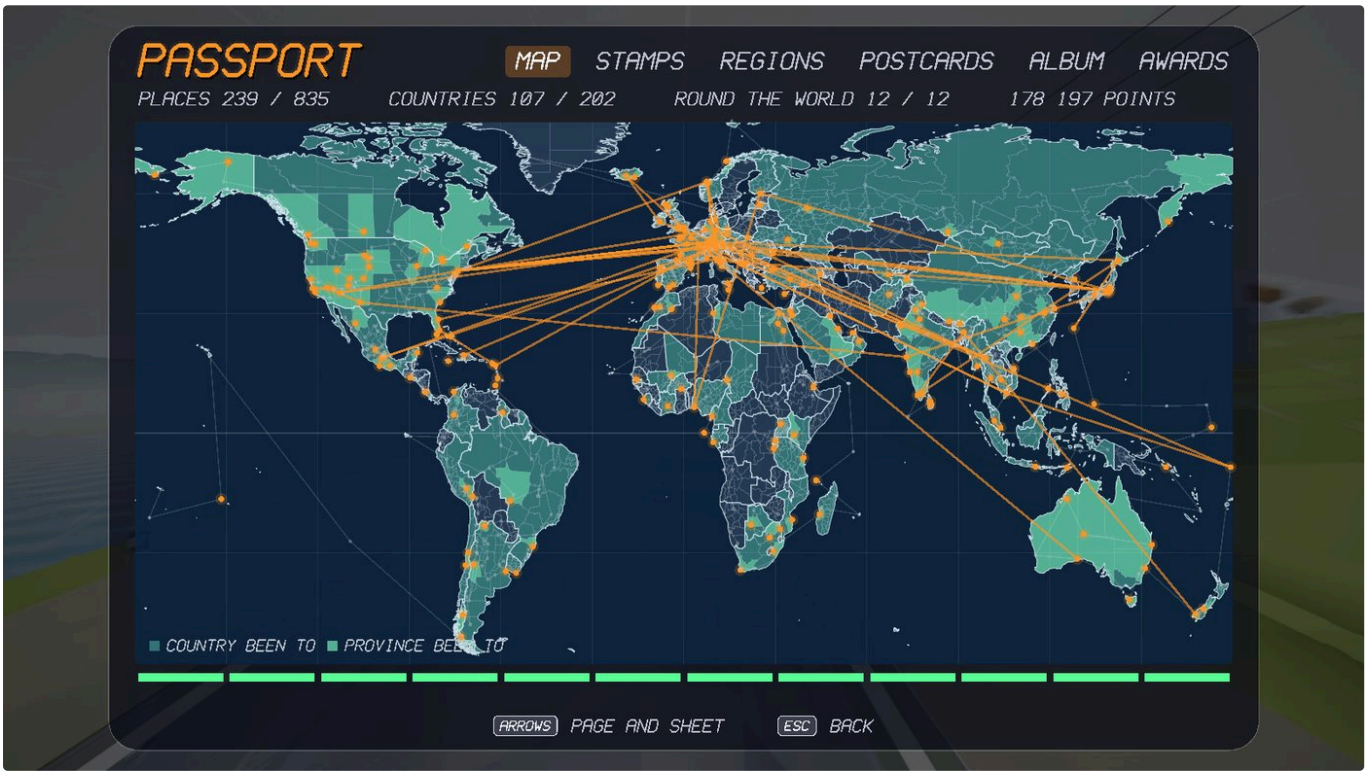
Car	What it is	Top speed	Engine	Yours
ARROW GT	the all-rounder, a V8	290 km/h	8 cylinders	from the start
SPARROW	light and revvy, a four-cylinder	259 km/h	4 cylinders	after 30 km
VORTEX	a V12, the fastest	317 km/h	12 cylinders	after 100 km or a title
KESTREL	a turbocharged rally hatchback, grippy on gravel and sand	274 km/h	4 cylinders	after 40 countries

The GARAGE dresses the car: fourteen paints, seven rim finishes and seven decals (twin stripes, a side stripe, a roundel, two tones, chequers, a globe) in any paint's colour. Career points from every run, every race and every new place and country pay for them; some keep conditions of their own (GLOBETROTTER needs a drive round the world, the globe 50 countries). The car carries the number plate of your HOME COUNTRY (at first your computer's region, NONE for no plate) in that country's style, and says what you type under NUMBER PLATE: up to eight letters, digits, spaces and dashes.

Passport and records

THE PASSPORT

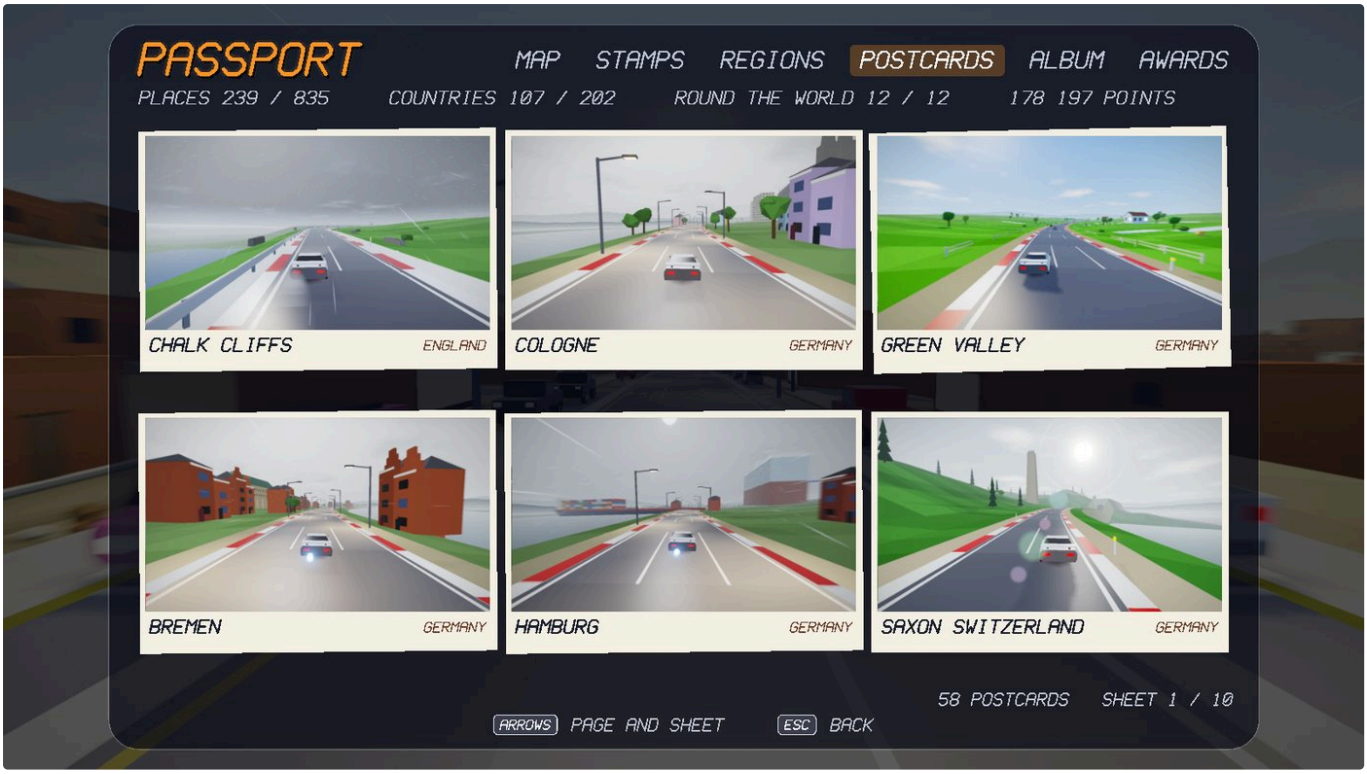
The passport shows the world tour on a map: the countries you have been to lit, every place a dot, the way you drove as a line, and the twelve slices of the globe as a bar (all twelve is a drive round the world). STAMPS has a stamp for every country; REGIONS shows the states and provinces of the big countries with how many you have seen; POSTCARDS has a card from every new place, taken in sight of its landmark; the ALBUM a photo of every landmark you have passed.



The map.



The stamps.



Postcards.

AWARDS

Award	How
First Stamp	Cross into a second country
Globetrotter	Drive through 25 countries
World Citizen	Drive through 100 countries
Every Flag	Every country of the tour
Across Europe	Every country of Europe
Across Africa	Every country of Africa
The Americas	From Alaska to Tierra del Fuego
The Silk Road and Beyond	Every country of South, Southeast and East Asia
Arabian Nights	Every country from Morocco to Oman
Down Under	Australia and New Zealand
Round the World	Cross every line of longitude
Getaway	Escape the police
Escape Artist	Escape the police ten times
Every Police	Escape the police of every livery
Caught Red-Handed	Get caught by the police
Good Samaritan	Take a hitchhiker where they wanted to go
Taxi Driver	Ten hitchhikers delivered
World Taxi	Hitchhikers from 20 countries
Make Way	Let ten ambulances and fire engines through
Local Knowledge	Beat a local hero to the checkpoint
Legend of the Roads	Beat 25 local heroes
Faster Than the Train	Beat a fast train to the end of its viaduct
Courier of the World	Ten courier jobs: parcels, organs, thieves
Legendary	Drive a legendary road from end to end
All the Legends	Drive every legendary road from end to end
Surf's Up	Aquaplane in a downpour

Postcard Collector	50 postcards in the album
Photographer	Photograph 25 landmarks
Long Haul	1000 km on the road
Road Warrior	10000 km on the road
Champion	Win a championship
Rally Legend	Finish every rally
Stamp Collector	Drive through 50 countries
Ambassador	Drive through 150 countries
Sightseer	Visit 100 places of the tour
Cartographer	Visit 400 places of the tour
Every Corner	Visit every place of the tour
Steppe and Taiga	Every country from the baltic to Mongolia
Island Hopper	Every country of the caribbean and the pacific
Twenty Thousand Leagues	Cross 10 seas
Wanderlust	25 stages of the random tour
Around the World By Chance	The random tour to every country
Party People	Be there for 3 festivals
All Year Round	Be there for every festival
Wish You Were Here	10 postcards in the album
Postmaster	200 postcards in the album
Say Cheese	Photograph a landmark
Photo Archive	Photograph 100 landmarks
Warmed Up	100 km on the road
Once Round the Earth	40075 km on the road
Marathon	100 km in one endless run
Six Figures	100000 points in one endless run
Slipstream	1000 overtakes
Traffic Surgeon	10000 overtakes
Most Wanted	Escape the police 50 times
Bus Driver	50 hitchhikers delivered
United Nations	Hitchhikers from 50 countries
Blue Light Friend	Let 50 ambulances and fire engines through
King of the Road	Beat 100 local heroes
Bullet Beater	Beat 10 fast trains
Special Delivery	50 courier jobs
The Classics	Drive 10 legendary roads from end to end
Water Skier	Aquaplane 25 times
Sideways	Drift 100 m in one go
Tyre Smoke	Drift 10 km in all
Tow	100 overtakes out of a slipstream
Hold Your Nerve	50 close calls with oncoming traffic
Daily Driver	The daily challenge on 7 days
Creature of Habit	The daily challenge on 30 days
On the Podium	Finish a championship in the top three
Dynasty	Win 3 championships
Rally Driver	Finish a rally
Every Journey	Finish every journey from a place to another



The awards in the passport.

RECORDS

The top ten with your initials, your total kilometres, titles, places and countries seen, and the best times of the rallies and journeys. The best drive on each track code is kept as a ghost.



The records.

Settings

Setting	What it does
PROFILE	BATTERY, BALANCED or QUALITY: resolution, shadows and frame rate in one go
DISPLAY	a window or fullscreen
3D RESOLUTION	the 3D picture at 50 to 200 % of the window; the text stays sharp
SHADOWS	the sun's shadows
VSYNC	wait for the display
FRAME LIMIT	a cap on the frame rate
DIFFICULTY	EASY, NORMAL or HARD: grip, clock, traffic and rivals
SCENERY	the world tour, the RANDOM TOUR to every country in random leaps, or ALL MIXED: every stage another biome
TIME OF DAY	each place's own light, the TOUR CLOCK, or DAWN, DAY, DUSK, NIGHT
WEATHER	weather fronts passing, or each place's own weather
SEASONS	AS TODAY, AS YOU TRAVEL (a year every 400 km) or OFF
POLICE	patrol cars that give chase
STEER ASSIST	OFF, LIGHT or STRONG
BRAKE ASSIST	brakes for bends and obstacles taken too fast
DRIFT	OFF, LOOK ONLY (the look and the points) or FULL (its feel too)
COLOUR BLIND	the screen's red and green become vermilion and sky blue
TEXT SIZE	100, 115 or 130 %
GEARS	automatic or manual
UNITS	km/h or mph
CAMERA	chase, far, bumper or cockpit
BLUR	the speed blur
VIBRATION	how strongly the gamepad rumbles
MUSIC	the music's volume
EFFECTS	the effects' volume
RADIO	the station, or AUTO
RESET SETTINGS	every setting back to its default (press twice)
ERASE PROGRESS	forget everything driven: records, places, unlocks, postcards, ghosts (press twice)

On the Steam Deck, or wherever frames matter, PROFILE switches the performance in one go: BATTERY (40 frames at 70 % resolution, no shadows), BALANCED (60 frames at 85 %, no shadows) or QUALITY (full resolution, shadows, as fast as the display). F3 shows the frame's cost in milliseconds, for the processor and the graphics card.

The settings and the records live in `%APPDATA%\Sunstreak\settings.ini` on Windows and in `~/.config/sunstreak` on Linux.

For tinkerers

THE DATA FILES

The world lives in the folder `data` as plain text, built into the program: every place with its light, land, road and what stands beside it; the countries with their road customs; the route of the tour; the seas; the landmarks and creatures in a small shape language; the festivals, the climate, the roadside. Set the environment variable `SUNSTREAK_DATA` to a copy of the folder to try changes without building; F5 reads the files again while the game runs, F6 opens the landmark viewer, every landmark and creature alone under its place's sky.

THE COMMAND LINE

Option	What it does
`--windowed | --fullscreen`	this start only
`--seed N`	the endless run's track (otherwise random)
`--biome N|ID`	the first stage's biome (on the world tour: where it starts)
`--scenery world|random|mixed|N|ID`	the endless run's scenery for this start (a biome's number or a place's ID such as TUSCANY: that one all the way)
`--play`	skip the title
`--championship`	start a championship
`--rally N`	start rally N (1..10)
`--journey N`	start journey N (1..)
`--daily`	start the daily challenge

`--action replay|photo|reload|save|siren|oncoming|overtaker T`	testing: the replay, the photo mode, reading the data again, saving the run, an ambulance or fire engine close by, three vehicles coming the other way close by, or further off with the last overtaking the others (ONCOMING TRAFFIC on, DENSE), after T seconds
`--resume`	continue the saved endless run
`--code XXXXX-XXXXX-XXXXX`	drive a Track Lab code
`--screen lab|records|code|garage|passport|stamps|regions|country|postcards|album|awards|rallies|viewer|settings`	open that screen
`--shape NAME`	the landmark the viewer opens on (with --screen country: the country)
`--shape-tour N`	with --screen viewer: photograph N models from there on, one picture each, saved as the --screenshot prefix followed by the model's name
`--camera 0..4`	chase, far, bumper or cockpit camera; 4 from above (screenshots)

`--find tunnel|bridge|landmark|landmark2|fork|sea|seain|seaout|ferry|jump|hitchhiker|hero|train|sky|summit|hump|gallery|runway`	start just before the first tunnel, bridge, landmark (or the second), fork, undersea portal (or inside, or before its far portal), ferry or crest
`--season spring|summer|autumn|winter`	that season (testing)
`--date MM-DD`	that date, for the seasons and the festivals (testing)
`--weather clear|rain|storm|fog|snow|sand|monsoon`	that weather all the way (testing)
`--time place|clock|dawn|day|dusk|night`	the time of day for this start
`--autopilot`	the car drives itself
`--travel`	travel: the world tour without a clock
`--look-back`	the camera looks back all the time (screenshots)
`--distance M`	start M metres down the road
`--oncoming N`	ONCOMING TRAFFIC for this run: 0 off, 1 light, 2 normal, 3 dense
`--size WxH`	window size for this start
`--quit-after S --screenshot F quit after S seconds, saving a screenshot`	
`--mute`	no sound (test runs)
`--stats`	start with the F3 performance overlay (CPU and GPU milliseconds per part of the frame)

`--bench F`	measure from six seconds into the drive, no vsync and no frame limit; with --quit-after, writes frame times, CPU parts and GPU passes to F, and how the drive went (crashes, scrapes, time off the road, drifting, slipstream)
`--audio-test F.wav`	render a scripted drive, each station and the adaptive layer offline, levels in F*.wav.txt

Credits and licences

Sunstreak is written in C++ on raylib (zlib licence), with SDL2 for the rumble (zlib licence). The outlines of the passport's map come from Natural Earth (public domain). The place names in their own scripts are traced from the Noto fonts (SIL Open Font License 1.1, in OFL-Noto.txt beside the game). The roadside of the places, their power lines, street lamps, guard rails and gravel roads, is derived from OpenStreetMap data, (c) OpenStreetMap contributors, under the Open Database License (ODbL); the derived data are available under the ODbL as well. The heights of the places come from ASTER GDEM, a product of METI and NASA, through OpenTopoData. The climate of the places comes from NASA POWER's thirty-year climatology (NASA Langley Research Center POWER Project); the smog through the year from Open-Meteo Air Quality (CC BY 4.0), which contains modified Copernicus Atmosphere Monitoring Service information; the road quality from the World Economic Forum's index; traffic and pollution from Numbeo's indices. CREDITS.txt beside the game lists them all. All drivers, heroes and hitchhikers are fictional.